Report Project Algorithms for Massive Datasets

Alessandro Crippa - 84583A

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1 Introduction

The project I have decided to develop is Project 3, on Link Analysis. Specifically, I chose to implement the PageRank algorithm on a user graph built from the New York Times Articles and Comments dataset.
I have implemented PageRank algorithm in two ways: the first using NumPy and Scipy's matrix operations and the second one using Apache Spark. Then, I have compared the two methods in terms of runtime, convergence and final results.
Finally, I have implemented Topic-Sensitive PageRank, calculated independently for some sections of articles and compared these specific rankings with the PageRank.

2 Dataset

The dataset I have used is the "New York Times Articles and Comments" (2020), published on Kaggle. This dataset was accessed in March 2026 and, since it is dynamically updated, results may slightly vary depending on the download time.
It consists of sveral CSV files, but in this project only two CSV files are considered: "nyt-articles-2020.csv", containing articles with columns "uniqueID" and "section"; "nyt-comments-2020.csv", that contains all the user comments with columns "userID", "articleID", "isAnonymous" and "status". For this project,

only the columns "userID", "articleID" and "section" have been used.

3 Data organization

The graph is constructed based on a co-commenting relationship: two users are connected if and only if they commented on at least one common article. This relationship is considered as an undirected relationship: however, since the PagerRank algorithm is defined for directed graphs, each undirected edge is considered as two reciprocal directed edges.
The graph is represented as a sparse adjacency matrix in CSR format (scipy.sparse .csr_matrix) of size $N \times N$, where N denotes the number of distinct users. As most pairs of users have never commented on the same article, a lot of the entries of the matrix are zero and the matrix is highly sparse. Storing a dense $N \times N$ matrix would require N^2 entries, which becomes very difficult for large values of N. Instead, the sparse matrix stores only the non-zero entries, significantly reducing memory usage.
An entry $A_{ij} > 0$ indicates that users i and j have commented on at least one article in common and the value in A_{ij} represents the number of articles that i and j have co-commented. However, once the graph is built, the matrix is binarized and the final value of A_{ij} is 1 if the two users have commented on at least one article in common, 0 otherwise.
To avoid generating an excessive number of edges for articles with many commenters, which would produce a large number of edges and degrade performance, a cap of "MAX USERS PER ARTICLE = 200" was imposed: this means that for article with more 200 commenters, only a random sample of 200 users is considered.
For the Apache Spark implementation, the matrix is converted to an RDD of (src, dst) edge pairs and reorganized into an adjacency list RDD of the form (node, [neighbors]).

4 Subsampling and pre-processing

Before building the graph, several pre-processing steps have been applied to ensure data quality.
The comments file is read in chunks of 100000 rows at time to avoid loading the entire file into memory rather than loading the entire file into memory at once. Within each chunk, two filters are applied: comments marked as anonymous ("isAnonymous == False") are removed since they can not be attributed to a specific users and comments with a "status" other than "approved" are excluded as they represent rejected or not yet published interactions. After concatenating all filtered chunks, duplicate ("userID, articleID") pairs are dropped: since what matters is whether two users commented on the same article, keeping duplicates would increase the edge weights without adding meaningful information.

When subsampling is enable, a random sample of SUBSAMPLE_USERS users is drawn from the total number of users. The sampling is random, meaning that every user has an equal probability of being selected regardless of how many articles they commented on. A fixed random seed ("seed == 42") is used to ensure that the same subset of users is selected across different executions, making the results fully reproducible.
The value of SUBSAMPLE_USERS acts as a global parameter: increasing it retains more users and produces a larger and denser graph at the cost of higher memory usage and longer execution times.
The filtered comments table is then merged with the articles table via a left join on "articleID" to retrieve the section column. This information is not relevant for PageRank algorithm, but is essential for the Topic-Sensitive PageRank, which requires knowing which section each article belongs to in order to define specific teleportation vectors. A final check removes any remaining rows with missing values in "userID" or "articleID".
An exploratory analysis has been carried out to understand the structure of the data.
The distribution of comments per article shows how many comments each article received, giving an indication of how much discussion it generates among users. The second distribution, the distribution of article per users, shows how many articles have been commented on by each user, reflecting how each user partecipates in the platform. Both distribution are right-skewed: most articles receive few comments (median 11, max 712) and most users commented on very few articles (median 1, max 2069), while a small number of articles and users dominate the dataset. The trend of these two distribution is shown in the figure 1 and 2. These statistic refer to an execution with SUBSAMPLE_USERS = 50000 and different values will generate different distribution.

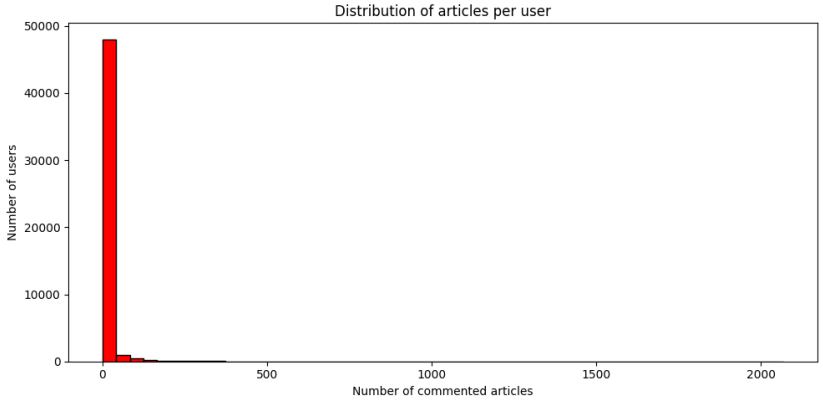

Figure 1: Distribution of articles per user.

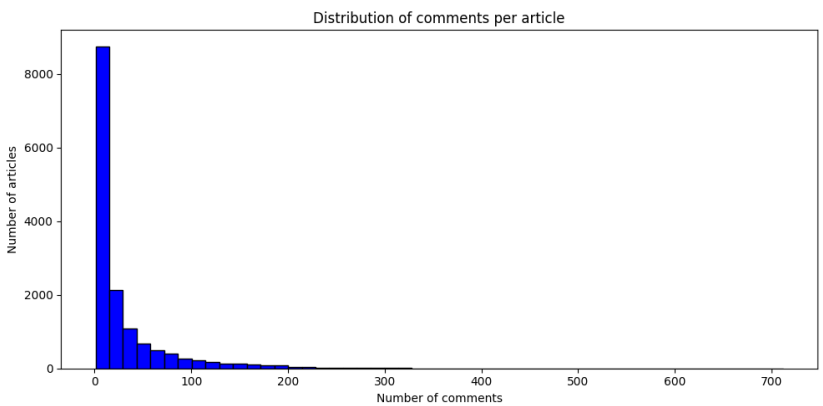

Figure 2: Distribution of comments per article.

5 Algorithms and implementations

5.1 PageRank

The PageRank algorithm was implemented from scratch using NumPy and SciPy sparse matrix operations and the implementations consists of two functions:

- build_transition_matrix(): it compute the column-normalized transition matrix M from the adjacency matrix A. Each column is divided by its sum, so that each column represents a probability distribution of transitioning from a node to its neighbors. Dead end nodes, that are nodes with no outgoing edges and they are identified by a zero column sum, are handled by temporarily setting their sum to 1 and leaving their column as all zero in M. This avoids division by zero while allowing their contribution to be redistributed at each iteration;

- pagerank(): it initializes the rank vector uniformly as $r_i = 1/N$ and iterates the following update rule:

$$r^{(t+1)} = \beta \cdot M \cdot r^{(t)} + \beta \cdot \frac{\sum_{i \in D} r_i^{(t)}}{N} \cdot \mathbf{1} + \frac{1-\beta}{N} \cdot \mathbf{1}$$

  where $\beta = 0.85$ is the probability that random surfer follows an edge at each iteration, D is set of dead end nodes and N the number of nodes. These three terms represent respectively: the rank propagated through the graph by following edges, the rank accumulated by dead end nodes redistributed uniformly to all nodes and the teleportation component that allows the random surfer to jump to any node with probability $1 - \beta$.
  Convergence is checked at each iteration by computing the L1 norm of the difference between consecutive rank vectors it is reached when this value falls below $\epsilon = 10^{-6}$ or after a maximum of 100 iterations.

5.2 PageRank with Spark

The Spark implementation follows the same mathematical formulation as the NumPy version but it distributes the computation across multiple partitions using PySpark Resiliant Distributed Datasets (RDDs).
The adjacency matrix A is converted to Coordinate format (COO format), which stores a sparse matrix as three array (row, cols and data) where each triple represents a non-zero entry of the matrix. This allow to easily extract the list of edges (src, dst) pairs, which are then parallelized into a Spark RDD. This is reorganized into an adjacency list RDD of the form (node, [neighbors]) using groupByKey(), deduplicated with distinct and cached in memory. The rank vector is represented as an RDD of (node, rank), initialized uniformly as $r_i = 1/N$. The values of N, β and the set of dead end nodes are distributed to all workers as broadcast variables, avoiding repeated data transfers across the network at each iteration.
Since Spark recomputes RDDs from scratch every time they are accessed, the adjacency list and rank RDDs are memorized with "MEMORY_AND_DISK", that means they are stored in memory or in disk if memory is insufficient. At each iteration, once the new rank RDD has been computed, the old one is unpersisted to free memory and avoid accumulation over many iterations.
At each iteration, the dead end node contribution and the teleportation term are computed as scalar values and combined into a single "base_rank". The rank propagation is then carried out in three steps: first, the adjacency list RDD is joined with the rank RDD; then each node distributes its rank equally among its neighbor; last, the received contributions are aggregated with combineKey() and scaled by β, adding "base_rank" to each result. A "leftOuterJoin()" with "all_nodes" ensures that nodes receiving no contributions are assigned "base_rank" rather being dropped.
Convergence is checked in the same way as it was done for PageRank.

5.3 Topic-Sensitive PageRank

The Topic-Sensitive PageRank extends the standard PageRank by replacing the uniform teleportation vector with a topic-specific one.
In this project, three topics are considered: "Opinion", "U.S." and "World". For each topic k, the set T_k of users who have commented on at least one article belonging to that section is identified. The teleportation vector is defined as:

$$\text{teleport}_i = \begin{cases} \frac{1}{|T_k|} & \text{if } i \in T_k \\ 0 & \text{otherwise} \end{cases}$$

The update rule becomes:

$$r^{(t+1)} = \beta \cdot M \cdot r^{(t)} + \beta \cdot (\sum_{i \in D} r_i^{(t)}) \cdot teleport + (1 - \beta) \cdot teleport$$

Compared to standard PageRank, both the teleportation component and the dead end nodes contribution are redistributed exclusively towards users active

in that specific topic.
The algorithm is run independently for each of three topics, using the same convergences criteria as the standard PageRank.

6 Scalability

The implemented solution is designed to scale to the full dataset through several concrete measures.
During data loading, the comments file is read in chunks of 100000 rows so that memory usage remains bounded regardless of file size and only the strictly necessary columns are loaded.The global variable "SUBSAMPLE" controls whether to work on a reduced subset or the full dataset: setting it to "False" is sufficient to process all data without any other code change.
During graph construction, the adjacency matrix is stored in CSR sparse format, whose memory usage grows with the number of edges. Edges are accumulated in buffers of 2000000 entries and added to matrix avoiding the need to hold all edges in memory at once. The cap on the number of users per article controls memory and computation by preventing the number of edges from growing quadratically for highly commented articles.

7 Experiments

Two experiments were conducted:

- PageRank comparison: it compares the NumPy and Spark PageRank implementations. Both run on the same graph with $\beta = 0.85, \epsilon = 10^{-6}$ and a maximum of 100 iterations. The goals are to measure execution time, number of iterations to convergence and consistency of the top 20 ranking (the 20 highest ranked users);

- Topic Sensitive and PageRank comparison: it runs independently the Topic-Sensitive PageRank for three topics ("Opinion", "U.S." and "World") and the goal is to compare the topic rankings against the normal implementation of PageRank to evaluate how the ranking changes based on the selected topic.

8 Results

The results of the two experiments are reported below:

- PageRank comparison: both implementation converged after 37 iterations producing identical top 20 rankings and scores. These results correspond to what was expected and they are shown is 3.
  The PageRank implemented using NumPy takes 6.63s to converge against 1497,59s (about 25 minutes) for Spark. This difference is expected because

Spark is designed for distributed systems but running it on a single machine makes it slower than a simple implementation of PageRank. The comparison graph of the convergence curves is shown in 4 which shows no difference as both implementation converge with the same number of iterations.

Rank	PR UserID	PR Score	PR_Spark UserID	PR_Spark Score	Same?
1	61986282	0.001118	61986282	0.001118	✓
2	2073520	0.000912	2073520	0.000912	✓
3	51905269	0.000901	51905269	0.000901	✓
4	75716722	0.000779	75716722	0.000779	✓
5	33668404	0.000762	33668404	0.000762	✓
6	80110445	0.000748	80110445	0.000748	✓
7	75215278	0.000713	75215278	0.000713	✓
8	51415176	0.000682	51415176	0.000682	✓
9	37617930	0.000672	37617930	0.000672	✓
10	62938502	0.000658	62938502	0.000658	✓
11	80370332	0.000653	80370332	0.000653	✓
12	48376006	0.000650	48376006	0.000650	✓
13	51087612	0.000636	51087612	0.000636	✓
14	23255351	0.000623	23255351	0.000623	✓
15	57353517	0.000612	57353517	0.000612	✓
16	87637841	0.000607	87637841	0.000607	✓
17	73444633	0.000595	73444633	0.000595	✓
18	69548808	0.000594	69548808	0.000594	✓
19	83218918	0.000590	83218918	0.000590	✓
20	60927884	0.000572	60927884	0.000572	✓

Figure 3: Comparison on rankings of PageRank and Spark PageRank.

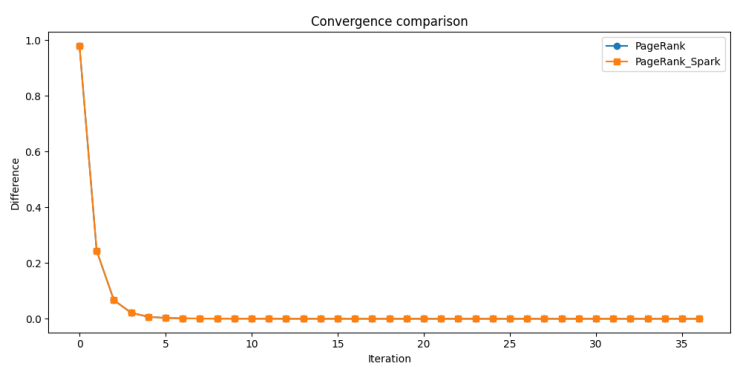

Figure 4: Comparison on convergence curve.

- Topic Sensitive and PageRank comparison: all the three topic reach convergence after 63 iterations, more than PageRank as the topic-specific teleportation slows convergence.
  The top 2 users retain the first two position across all three topics suggesting they are globally dominant regardless of topic. As the figure 5

shows, based on the topic analyzed, there may be changes in position or even disappearance of users from the top 20.

Rank	UserID	Score	PR Rank	TSPR_Opinion	TSPR_U.S.	TSPR_World
1	61986282	0.001118	1	1	1	1
2	2073520	0.000912	2	2	2	2
3	51905269	0.000901	3	3	3	3
4	75716722	0.000779	4	7	5	4
5	33668404	0.000762	5	6	4	5
6	80110445	0.000748	6	5	6	6
7	75215278	0.000713	7	8	9	11
8	51415176	0.000682	8	9	8	9
9	37617930	0.000672	9	4	11	10
10	62938502	0.000658	10	10	10	12
11	80370332	0.000653	11	13	12	15
12	48376006	0.000650	12	14	7	7
13	51087612	0.000636	13	12	17	16
14	23255351	0.000623	14	39	25	21
15	57353517	0.000612	15	11	18	17
16	87637841	0.000607	16	47	45	27
17	73444633	0.000595	17	20	15	8
18	69548808	0.000594	18	29	13	14
19	83218918	0.000590	19	25	16	29
20	60927884	0.000572	20	19	21	13

Figure 5: Comparison on rankings of Topic-Sensitive and PageRank.